SLIC INDEX · PROJECT TECHNICAL PAPER

The SLIC Index

Publishing an Auditable City Index: Methods and Lessons from Six Months in Public

“Every city ranking is a lie. Here’s ours — public methodology, public data, reproducible code, no paid placement.”

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163 cities · 158 ranked · 5 watchlist · 5 pillars · 22 scored indicators + 3 diagnostics

Score model: slic-v3.4.0 · Tier policy: public-tier-v1.3.0 · Published snapshot: 2026-10-29T13:00:57.991Z

Open Data & Interactive Engine: https://slic.nonarkara.org · Code Repository: github.com/Nonarkara/SLIC-Index

The Smart and Liveable Cities (SLIC) Index

Methodology Technical Paper · Score model slic-v3.4.0

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Abstract

The Smart and Liveable Cities (SLIC) Index is an open, reproducible model of urban liveability. Drawing upon the Adjusted Mazziotta-Pareto Index (AMPI), SLIC penalizes cross-pillar imbalance ($\mu - \sigma^2/\mu$). This reduces compensation between pillars; it does not eliminate it. Public tier floors and country quotas are separate from the numerical score. SLIC also applies declared coverage penalties and withholds a ranking where coverage is insufficient. The published snapshot contains 163 cities, including 158 ranked cities and 5 watchlist entries, across five pillars (Growth 25%, Viability 22%, Capability 18%, Community 15%, Creative 20%). Inputs include city measurements, national proxies, derived quantities and composites. Matching an input to the source register establishes traceability, not independent empirical validation. Registered source dates are not independently verified observation periods. The results describe this declared model and its limitations, not a certification of a city's quality or institutional endorsement. This retrospective examines what public operation revealed about maintaining that agreement between data, code and explanation. Of 4,193 raw or component inputs, 4,187 match register rows, 6 are declared exceptions and 180 lack matched dates. Correction records identify geographic-scope mislabelling, inconsistent hash canonicalization and misleading displayed arithmetic. The principal lesson is procedural: public reproducibility requires explicit treatment of proxies, exceptions and corrections, not merely publication of an aggregation formula. No causal claim about urban outcomes or reader behavior is made.

Keywords: Smart Cities, Urban Liveability, Adjusted Mazziotta-Pareto Index (AMPI), Imbalance Penalty, Urban Informatics.
JEL Classification: R00, R11, I31, H41, O18

Snapshot fact	Value
Cities in public file	163
Ranked cities	158
Watchlist cities	5
Scored metric lines	22
Visible non-scoring diagnostics	3

Publication as a method: a retrospective

Study design and chronology

This is a retrospective technical case study of a publicly accessible ranking instrument, not an evaluation of city-policy effects. The project's chronology records a March 2026 Taipei launch of V2. The six-month framing describes the subsequent period of public development; it is not an exact anniversary, a six-month visitor panel or a longitudinal sample of city outcomes. The numerical analysis below refers to the frozen publication dated 2026-10-29T13:00:57.991Z, not to observations all collected on that date. The unit of analysis is the publication pipeline: its inputs, declared transformations, interfaces and correction records.

What the records establish

The source-register audit accounts for 4,193 non-null raw inputs or component inputs. It matches 4,187 to register rows and declares 6 exemptions; 4,013 have registered dates and 180 do not. These counts are traceability results, not counts of independent city measurements. National proxies and repeated country values are not independent observations. The exemptions include commercial investment measures and social-mobility measures that cannot be reread from the cited public pages. Those exceptions also expose construct differences: a Gini coefficient and a social-mobility index do not measure the same thing, nor do national FDI and city venture-capital deal density. Register completeness does not resolve that comparability problem.

Public reach, with an incomplete denominator

The archived V1/V2 visitor snapshot contains 1,742 records between 2026-03-10 and 2026-10-19: 106 days with records within a 224-day calendar span. The archive may include repeat visits and automated traffic. Missing days are unobserved, not zeros, and no missing traffic is imputed. The replacement first-party counter measures acknowledged browser-tab sessions and is reported separately. Neither source establishes unique people, endorsement, methodological understanding or policy uptake. Worldwide access is evidence of distribution, not evidence that the instrument improved a decision.

Corrections as findings about the instrument

The October reconciliation documented country-level proxies labelled as city evidence. Correcting those labels did not itself change raw values or rankings. Later source adoption can change scores and therefore requires a separate publication, not a cosmetic metadata patch. A second failure concerned canonical hashes: scripts using different object-key orders disagreed about identical data. A shared canonical payload now makes those checks reproducible, but unsigned hashes prove internal consistency, not source authenticity. A third failure was visible arithmetic: rounded pillar terms could disagree with a separately rounded mean. The repair retained the original tolerance, increased intermediate display precision, and made missing-pillar renormalization and the zero floor explicit. These incidents show why an open formula alone is insufficient: provenance labels, executable code and what a reader can actually recompute must agree.

Interpretation and limits

SLIC's contribution in this case is a contestable publication workflow: readers can inspect assumptions and explore alternative weights. That is a design affordance, not a measured behavioral result. No controlled comparison, representative user study or independent validation of all inputs is reported here. Changing weights does not repair incompatible constructs, stale observation periods, national-to-city transfer or uneven source coverage. The imbalance penalty reduces compensation between dimensions; it cannot establish that the chosen dimensions are sufficient. Country quotas and tier floors express editorial choices rather than statistical discoveries. The next evaluation should independently audit a stratified input sample, distinguish source dates from observation periods, test rankings under defensible proxy substitutions and normalization choices, and observe whether readers interpret the score and missingness correctly.

Evidence available for replication

The archived visitor aggregates are in src/data/traffic-history.json; current traceability counts are in src/data/evidenceSummary.json and the downloadable input register. The source registers, declared exemptions and aliases are in data/verified_sources/. The reconciliation and release records are docs/evidence-reconciliation-2026-10-25.md, docs/inclusion-evidence-reconciliation-2026-10-28.md and docs/security/release-recovery-2026-10-29.md. The public repository supplies the scoring implementation and

regression tests. These records permit inspection of the reported corrections; they do not substitute for checking the underlying providers or their redistribution terms.

Acknowledgements

The authors thank Mr. Vichate Tantiwanich, Chairman of the Board of Smart and Livable City Co., Ltd., for proposing the original idea for this benchmark. His question helped frame the project: what matters in everyday urban life, and whose priorities does a city ranking represent? Rather than treating the preferences of a particular social class as a universal standard, the project makes its choices of indicators and weights explicit and open to challenge. This acknowledgement credits the originating idea; responsibility for the methods, interpretations and remaining errors rests with the authors. It does not imply his endorsement of individual city rankings.

Role reference: Asia Aviation, biography of Mr. Vichate Tantiwanich, https://www.aavplc.com/en/leadership/thai-airasia/board-of-directors/175/mr-vichate-tantiwanich (accessed 4 November 2026). The account of the project's originating idea is the authors' acknowledgement.

1. Published method

SLIC has five public pillars. Within each pillar, observed scored metrics are combined by weighted mean. Across pillars, the final score uses a weighted AMPI, so cross-pillar imbalance is punished mathematically rather than rhetorically.

```
normalize(x, p05, p95, positive) = 100 * clamp((winsor(x) - p05) / (p95 - p05), 0, 1)
normalize(x, p05, p95, negative) = 100 * clamp((p95 - winsor(x)) / (p95 - p05), 0, 1)

P_p(c) = sum observed alpha_(p,m) * q_m(c) / sum observed alpha_(p,m)

O(c) = pillars with observed scores; w = (25, 22, 18, 15, 20)
W(c) = sum over O(c) of w_p
mu(c) = sum over O(c) of w_p * P_p(c) / W(c)
var(c) = sum over O(c) of w_p * (P_p(c)-mu(c))^2 / W(c)
No observed pillars: no numeric mean or AMPI
AMPI(c) = clamp(mu(c) - var(c)/max(mu(c), 1e-10), 0, 100)
All-zero observed scores: mu = var = AMPI = 0
SLIC(c) = max(0, AMPI(c) - coverage_penalty(c))
```

The final published score is therefore **not** a simple weighted average of the displayed pillar scores. That distinction matters: a city with one catastrophic pillar is supposed to score worse than a city with the same weighted mean but more even performance.

2. What is scored

The current public model scores **22** metric lines and keeps **3** additional metric lines visible as diagnostics only. Visibility and scoring are intentionally separated.

Growth (25)

Metric	Weight	Status	Method note
Tax-adjusted PPP disposable income	8	Scored	Residual money after essentials, converted once through the PPP private-consumption factor.
Housing burden	5	Scored	Purchase price per sq ft (USD, Numbeo 2024-25). Higher price = higher burden = lower score. Direction: negative.
Household debt burden	2	Scored	Debt relative to income, with fallback when city debt evidence is thin.
Working time pressure	4	Scored	Higher work burden scores worse.
Suicide and severe mental strain	4	Scored	Public-health strain proxy; higher severe-strain burden scores worse. The declared weight follows the published metric catalog.
Macroeconomic stress (Hanke Misery Index)	2	Scored	Hanke Annual Misery Index 2025: 2×unemployment + inflation + bank-lending-rate – real GDP/capita growth. Country-level. Lower score = less macro stress = higher SLIC score.

Metric	Weight	Status	Method note
Economic growth momentum	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Viability (22)

Metric	Weight	Status	Method note
Personal safety	5	Scored	Harm and victimization outcomes, not surveillance theatre.
Transit access and commute burden	5	Scored	Better access and lower burden score better.
Clean air	4	Scored	Higher pollution exposure scores worse.
Water, sanitation, and utility reliability	4	Scored	Safer access and higher reliability score better.
Digital infrastructure	4	Scored	Broadband quality and affordability.
Climate and sunlight livability	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Capability (18)

Metric	Weight	Status	Method note
Healthcare quality	8	Scored	Effective-care quality and access.
Education quality	6	Scored	Learning quality and skills formation.
Equal opportunity and distributional fairness	4	Scored	Composite scored metric: 70% equal-opportunity evidence, 30% reversed Gini context.

Community (15)

Metric	Weight	Status	Method note
Hospitality and belonging	4	Scored	Do people feel welcome in practice?
Tolerance and pluralism	4	Scored	Composite scored metric: 40% Equaldex, 30% Freedom House, 30% reversed hate-crime or civil-liberties proxy.
Cultural, historic, and public-life vitality	3	Scored	Everyday public-life texture; visitor demand is contextual, not an automatic bonus.
Civic freedom and dignity	4	Scored	Composite: 0.4 × HRMI Empowerment + 0.3 × Freedom House FIW + 0.3 × V-Dem Liberal Democracy Index (×100). Higher = more civic freedom. This measures a different construct from macroeconomic stress.
Birth-rate optimism	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Creative (20)

Metric	Weight	Status	Method note
Entrepreneurial dynamism	6	Scored	Startup formation and productive entry.
Innovation and research intensity	5	Scored	Research depth and knowledge production.
Economic vitality and productive context	5	Scored	Composite scored metric: 50% investment signal, 30% GDP per capita PPP, 20% GDP growth.
Administrative and investment friction	4	Scored	Higher time-cost and licensing burden score worse.

3. Composite metrics

Three scored terms are composites. When one component is missing, the composite is reweighted over the observed component weights only; nothing is silently imputed.

```
EqualOpportunityFairness = 0.7 * EqualOpportunity + 0.3 * ReverseGini
TolerancePluralism      = 0.4 * Equaldex + 0.3 * FreedomHouse + 0.3 * ReverseHateCrime
EconomicVitality        = 0.5 * InvestmentSignal + 0.3 * GDPperCapitaPPP + 0.2 * GDPGrowth
```

4. Coverage, watchlist logic, and ranking protocol

```
cov_direct = 1 if observed else 0
cov_composite = observed component weight / total component weight

Cov_p(c) = sum alpha_(p,m) * cov_m(c) / sum alpha_(p,m)
Cov(c)   = (25 Cov_pressure + 22 Cov_viability + 18 Cov_capability +
            15 Cov_community + 20 Cov_creative) / 100
```

Coverage grade	Rule	Penalty	Rank status
A	Cov >= 0.75	0	Ranked
B	0.50 <= Cov < 0.75	5	Ranked
C	0.35 <= Cov < 0.50	15	Ranked
Watchlist	Cov < 0.35 or manual watchlist override	0	Watchlist

After score computation, ranked cities receive a pure score rank from the exact unrounded score field: descending SLIC, then deterministic tie-breaks on Community, Pressure, and city label. Alpha, Beta, and Gamma are then computed as a separate public overlay. The overlay allows one city per country across all three tiers, except Taiwan may hold 2 public-tier seats and Japan may hold 2 public-tier seats; Alpha requires Community >= 40, Pressure >= 40, and coverage grade A, so missing stress inputs cannot be converted into a top-shelf public claim; Europe is capped at 2 Alpha seats, Oceania capped at 0, South Korea capped at 1, Japan capped at 1, Israel excluded from Alpha by country, and the editorial city list (Tokyo, Hong Kong, Sydney, Vancouver, Los Angeles, Adelaide, Honolulu, San Francisco, Melbourne, San Diego, Brisbane, London, Amsterdam, Athens, Prague, Košice, Kosice) also excluded from Alpha — these cities are barred from Alpha because their median-resident cost-of-living puts them outside SLIC's positive showcase; Beta raises the floors to Community >= 45 and Pressure >= 45; Gamma fills from the remaining ranked cities.

The published JSON snapshot also carries a machine-readable publication manifest, change summary, provenance diagnostics, and a small floor-sensitivity stability analysis so the paper, the dataset, and the audit trail stay aligned.

Live example: Singapore is currently pure rank **#33** with SLIC **63.3**, qualifying for the **Alpha** tier (slot 7) with world-leading Viability (90.4) and Capability (94.1) under the single-city-per-country quota. Bangkok is pure rank **#66** with SLIC **53.5**, demonstrating strong Viability (76.6) and Community (68.4) anchored in Southeast Asia's urban core.

5. Worked example from the live dataset

Example city: **Kaohsiung**. This walkthrough uses the published row in the live dataset, not a fictional demonstration.

```
Negative-direction normalization example (personal safety)
raw = 0.8
p05 = 0.23
p95 = 22.63
score = 100 * (22.63 - 0.8) / (22.63 - 0.23)
score = 97.5

Growth weighted mean
Growth = (8x81.5 + 5x81.6 + 4x41.4 + 4x27.5 + 2x100.0) / 23
Growth = 66.8

Cross-pillar AMPI
mu  = 68.862
var = 362.402
AMPI = 68.862 - 362.402 / 68.862
AMPI = 63.599
Coverage = 0.8 -> grade A -> penalty 0
```

```
Published SLIC = 63.6
```

The important methodological point is visible in the arithmetic: the weighted pillar mean for this city is higher than the final score because AMPI makes uneven pillar structure costly.

6. Data profile of the current public file

Measure	Value
City / metro metric share	12.9%
National / international metric share	61.0%
Composite metric share	13.3%
Derived metric share	12.9%
Grade A cities	138
Grade B cities	5
Grade C cities	19
Watchlist cities	5

These shares are computed directly from non-null metric lines in the published JSON. They are included here so the paper does not rely on invented methodology graphics or unsupported source-mix claims.

7. Published ranked cities

The table below is the ranked slice of the current published dataset. Watchlist cities are excluded from this list by definition.

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
1	Montreal	Canada	73.9	69.7	94.9	84.0	82.1	54.1	A
2	Gothenburg	Sweden	72.5	72.9	100.0	85.5	75.4	63.3	B
3	Bergen	Norway	72.1	80.4	100.0	96.8	79.5	48.8	B
4	Raleigh	United States	71.8	77.6	81.6	74.4	64.0	61.5	A
5	Eindhoven	Netherlands	69.6	68.8	98.2	87.3	75.8	42.8	A
6	Ottawa	Canada	69.4	65.6	94.9	84.0	83.4	44.2	A
7	Minneapolis	United States	68.9	75.6	76.4	74.4	56.3	61.5	A
8	Pittsburgh	United States	68.3	78.6	73.3	74.4	61.1	55.3	A
9	Melbourne	Australia	67.6	49.9	92.7	92.4	69.6	60.5	A
10	Prague	Czechia	67.6	68.9	90.6	71.1	87.5	42.6	A
11	Perth	Australia	67.4	56.3	92.7	92.4	69.6	51.2	A
12	Copenhagen	Denmark	67.1	43.7	98.4	81.6	91.0	61.5	A
13	Graz	Austria	66.9	65.5	90.1	83.1	77.6	41.4	A
14	Toronto	Canada	66.9	50.4	94.9	84.0	80.9	54.1	A
15	Brisbane	Australia	66.6	51.8	92.7	92.4	69.6	54.4	A
16	New York	United States	66.4	43.8	85.4	82.2	70.5	74.5	A
17	Adelaide	Australia	66.4	55.5	92.7	92.4	69.6	49.2	A
18	Braga	Portugal	65.5	64.6	96.7	74.0	69.3	43.8	A
19	Chicago	United States	64.9	64.3	66.8	74.4	59.0	61.5	A
20	London	United Kingdom	64.9	54.8	82.8	69.4	71.2	57.7	A
21	Jeju City	South Korea	64.9	56.4	98.1	92.3	55.7	50.6	A
22	Helsinki	Finland	64.9	47.6	93.5	94.5	71.2	52.9	A
23	Sydney	Australia	64.4	40.7	92.7	92.4	69.6	63.9	A
24	Vancouver	Canada	64.4	45.1	94.9	84.0	79.0	54.1	A
25	Incheon	South Korea	64.4	55.9	93.2	92.3	55.7	50.6	A
26	Christchurch	New Zealand	64.0	56.6	94.3	85.7	71.9	41.9	A
27	Wellington	New Zealand	64.0	50.1	94.3	85.7	71.9	48.8	A
28	Lyon	France	63.9	56.3	98.4	71.8	75.4	45.1	A
29	Dunedin	New Zealand	63.8	61.4	94.3	85.7	71.9	37.0	A
30	Tallinn	Estonia	63.8	56.4	92.4	67.6	57.5	57.8	A
31	Kaohsiung	Taiwan	63.6	66.8	86.5	90.6	62.9	37.0	A
32	Porto	Portugal	63.5	58.2	96.7	74.0	69.3	43.8	A
33	Singapore	Singapore	63.3	44.1	90.4	94.1	47.4	73.3	A
34	Busan	South Korea	63.0	51.9	95.9	92.3	55.7	50.6	A
35	Hobart	Australia	62.9	50.2	92.7	92.4	69.6	45.1	A
36	Auckland	New Zealand	62.8	44.4	94.3	85.7	71.9	52.9	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
37	Suwon	South Korea	62.7	51.4	93.1	92.3	55.9	50.6	A
38	Fukuoka	Japan	62.3	60.0	95.8	74.6	60.8	42.9	A
39	Manama	Bahrain	62.2	72.4	66.9	57.8	45.2	67.5	A
40	Valparaiso	Chile	61.6	63.2	76.4	67.5	68.3	43.0	A
41	Santiago	Chile	61.4	57.4	72.9	77.6	68.3	44.6	A
42	Sapporo	Japan	61.1	62.1	95.8	74.6	60.8	37.9	A
43	Milan	Italy	61.1	77.8	85.8	71.7	65.9	29.4	A
44	Hiroshima	Japan	60.8	61.2	95.8	74.6	60.8	37.9	A
45	Taipei	Taiwan	60.8	45.4	89.5	77.3	62.9	53.7	A
46	Kobe	Japan	60.8	61.0	95.8	74.6	60.8	37.9	A
47	Zurich	Switzerland	60.5	40.4	98.2	78.1	82.2	49.4	A
48	Venice	Italy	59.9	59.1	84.7	74.0	69.5	35.6	A
49	Amsterdam	Netherlands	59.7	42.9	98.2	87.3	75.8	42.8	A
50	Dubai	United Arab Emirates	59.7	75.7	72.4	76.7	37.8	46.2	A
51	Cork	Ireland	59.7	88.1	100.0	86.7	91.1	37.4	C
52	Abu Dhabi	United Arab Emirates	59.6	76.6	70.8	76.7	37.8	46.2	A
53	Vienna	Austria	59.5	45.9	90.1	83.1	75.7	41.4	A
54	Tokyo	Japan	59.5	49.0	94.2	70.5	73.9	42.1	A
55	Bratislava	Slovakia	58.5	60.4	91.2	65.3	64.8	35.3	A
56	Brno	Czechia	58.1	73.5	92.5	100.0	87.3	42.4	C
57	Antwerp	Belgium	57.9	73.2	100.0	95.6	88.5	40.4	C
58	Torun	Poland	57.9	67.2	77.3	77.1	56.1	31.9	A
59	Katowice	Poland	57.1	65.8	73.7	77.1	56.1	31.9	A
60	Gdansk	Poland	57.0	61.0	83.4	77.1	56.1	31.9	A
61	Bologna	Italy	56.9	59.7	87.9	71.7	65.9	29.4	A
62	Port Louis	Mauritius	55.3	59.1	83.6	48.6	51.2	45.4	A
63	San Juan	Puerto Rico	55.3	59.3	68.8	68.3	67.0	45.8	B
64	Bucharest	Romania	55.1	61.2	85.8	61.0	51.6	34.8	A
65	Krakow	Poland	54.7	56.1	74.6	77.1	56.1	31.9	A
66	Bangkok	Thailand	53.5	42.6	76.6	68.2	68.4	38.5	A
67	Paris	France	53.3	33.1	98.4	71.8	72.4	45.1	A
68	Ljubljana	Slovenia	53.1	69.3	94.4	100.0	86.4	33.2	C
69	Montevideo	Uruguay	51.8	53.0	72.0	72.2	71.6	24.4	A
70	Zagreb	Croatia	51.3	67.1	85.0	82.2	72.3	42.3	C
71	San Jose	Costa Rica	51.2	55.3	55.9	56.3	61.0	35.9	A
72	Budapest	Hungary	50.3	48.3	94.7	64.9	62.8	26.2	A
73	Melaka	Malaysia	50.2	67.0	75.3	49.1	43.3	31.5	A
74	Kota Kinabalu	Malaysia	50.0	65.6	77.9	49.1	43.1	31.5	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
75	Kuching	Malaysia	49.8	65.7	72.0	49.1	43.8	31.5	A
76	Riga	Latvia	49.3	65.4	100.0	66.5	76.5	40.3	C
77	Male	Maldives	49.2	54.6	80.7	50.8	41.7	35.4	A
78	George Town	Malaysia	49.2	62.7	71.1	49.1	43.8	31.5	A
79	Doha	Qatar	49.1	67.2	56.5	58.1	33.4	37.6	A
80	Panama City	Panama	49.1	71.5	61.2	49.6	52.6	26.9	A
81	Vilnius	Lithuania	48.9	68.5	100.0	58.4	77.9	41.4	C
82	Kuala Lumpur	Malaysia	48.0	56.1	72.4	49.1	43.8	31.5	A
83	Jeddah	Saudi Arabia	47.8	95.5	66.7	78.9	24.9	24.1	A
84	Khobar	Saudi Arabia	47.8	94.7	66.7	78.9	24.9	24.1	A
85	Riyadh	Saudi Arabia	47.8	94.5	66.7	78.9	24.9	24.1	A
86	Valencia	Spain	47.5	67.6	100.0	68.9	86.6	30.4	C
87	Guangzhou	China	47.0	53.2	91.0	74.6	20.6	37.1	A
88	Tianjin	China	46.8	55.7	72.6	74.6	20.6	37.1	A
89	Curitiba	Brazil	46.8	57.2	58.0	53.9	47.7	27.6	A
90	Florianopolis	Brazil	46.6	56.2	58.0	53.9	47.7	27.6	A
91	Belgrade	Serbia	46.5	48.9	72.0	59.8	39.3	30.3	A
92	Salalah	Oman	46.1	80.0	71.7	42.0	32.3	30.6	A
93	Cordoba	Argentina	46.1	62.7	78.7	73.1	56.5	10.2	A
94	Hat Yai	Thailand	46.0	53.3	66.7	62.2	46.7	22.8	A
95	Sao Paulo	Brazil	45.8	52.3	58.0	53.9	47.7	27.6	A
96	Shenzhen	China	45.5	49.2	92.7	74.6	20.6	37.1	A
97	Kuwait City	Kuwait	45.3	93.9	52.9	65.0	41.3	19.9	A
98	Chengdu	China	45.2	48.7	77.0	74.6	21.1	37.1	A
99	Chongqing	China	45.2	49.1	75.2	74.6	21.0	37.1	A
100	Nanjing	China	44.6	66.0	75.3	61.3	10.1	39.4	A
101	Hangzhou	China	44.4	45.6	84.0	74.6	21.1	37.1	A
102	Phuket	Thailand	44.4	46.9	68.0	62.2	46.7	22.8	A
103	Muscat	Oman	44.4	74.8	60.3	42.0	32.3	30.6	A
104	Buenos Aires	Argentina	44.3	55.8	78.7	73.1	56.5	10.2	A
105	Shanghai	China	44.1	45.3	85.2	74.6	20.6	37.1	A
106	Suva	Fiji	43.5	55.3	51.2	57.0	40.7	25.1	A
107	Chiang Mai	Thailand	43.1	51.9	49.7	62.2	46.7	22.8	A
108	Thimphu	Bhutan	42.7	50.9	46.2	37.3	47.4	34.7	A
109	Rabat	Morocco	42.0	51.8	66.7	34.1	27.1	41.2	A
110	Casablanca	Morocco	41.7	51.4	62.3	34.1	27.1	41.2	A
111	Tbilisi	Georgia	41.1	55.7	77.6	66.9	47.5	43.1	C
112	Arequipa	Peru	39.5	62.7	40.8	38.5	45.8	24.7	A
113	Aqaba	Jordan	39.1	48.5	62.5	46.5	12.7	40.8	A
114	Bogota	Colombia	39.1	55.2	39.2	50.7	50.3	19.6	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
115	Amman	Jordan	38.4	46.7	59.0	46.5	12.7	40.8	A
116	Lima	Peru	38.3	59.5	37.3	38.5	45.8	24.7	A
117	Medellin	Colombia	38.3	58.3	35.2	50.7	50.3	19.6	A
118	Nizhny Novgorod	Russia	36.9	67.0	70.8	65.6	15.6	15.5	A
119	Bengaluru	India	36.5	46.8	54.5	46.0	41.7	14.6	A
120	Hyderabad	India	36.3	47.0	52.7	46.0	41.7	14.6	A
121	Makati	Philippines	36.2	56.9	45.7	38.9	38.8	17.5	A
122	Cebu City	Philippines	36.2	59.5	44.8	38.9	38.8	17.5	A
123	Pune	India	35.1	46.5	44.8	46.0	41.7	14.6	A
124	Chandigarh	India	34.9	46.0	44.1	46.0	42.1	14.6	A
125	Da Nang	Vietnam	34.7	58.3	77.6	49.6	26.3	20.8	B
126	Munich	Germany	34.5	30.3	100.0	72.9	80.5	36.2	C
127	Gaborone	Botswana	33.3	52.4	45.3	6.7	53.9	39.0	A
128	Merida	Mexico	33.2	60.1	29.3	48.7	48.0	14.1	A
129	Surabaya	Indonesia	33.2	67.0	36.2	39.1	33.6	17.8	A
130	Windhoek	Namibia	33.1	44.0	51.9	16.5	50.4	25.4	A
131	Santo Domingo	Dominican Republic	32.8	55.2	88.8	46.3	52.8	28.2	C
132	Jakarta	Indonesia	32.7	63.7	34.5	39.1	33.6	17.8	A
133	Port Vila	Vanuatu	32.5	43.3	100.0	73.3	63.7	21.9	C
134	Cape Town	South Africa	32.3	31.8	35.1	26.3	54.7	29.8	A
135	Moscow	Russia	32.1	44.8	71.2	65.6	15.6	15.5	A
136	Mexico City	Mexico	31.8	52.3	27.1	48.7	48.0	14.1	A
137	Guadalajara	Mexico	31.7	57.8	25.8	48.7	48.0	14.1	A
138	Johannesburg	South Africa	31.6	33.0	31.1	26.3	54.7	29.8	A
139	Kandy	Sri Lanka	31.5	48.3	49.4	34.2	35.9	11.7	A
140	Colombo	Sri Lanka	31.1	45.8	48.5	34.2	35.9	11.7	A
141	Phnom Penh	Cambodia	30.3	54.6	55.1	18.2	28.2	36.3	B
142	Kigali	Rwanda	29.3	59.5	47.3	14.2	27.9	24.8	A
143	Pokhara	Nepal	28.4	48.9	36.5	22.4	34.2	16.6	A
144	Kathmandu	Nepal	27.0	47.8	30.3	22.4	34.2	16.6	A
145	Nairobi	Kenya	26.5	54.2	53.9	13.0	35.2	14.5	A
146	Dhaka	Bangladesh	25.6	51.4	36.9	19.8	25.6	15.7	A
147	Chattogram	Bangladesh	25.6	52.9	36.9	19.8	25.6	15.7	A
148	Accra	Ghana	21.8	59.0	29.0	8.1	41.5	17.8	A
149	Kumasi	Ghana	21.1	60.7	27.5	8.1	41.5	17.8	A
150	Dakar	Senegal	20.9	44.4	12.6	13.0	39.0	26.7	A
151	Asuncion	Paraguay	18.0	52.9	96.3	25.3	51.8	14.9	C
152	Dar es Salaam	Tanzania	17.1	42.4	70.1	40.2	33.4	12.7	C
153	Islamabad	Pakistan	16.2	46.3	25.0	18.8	22.0	4.4	A
154	Karachi	Pakistan	16.2	46.5	25.0	18.8	22.0	4.4	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
155	Lahore	Pakistan	16.2	46.8	25.0	18.8	22.0	4.4	A
156	Cuenca	Ecuador	12.2	51.5	100.0	21.2	56.8	8.0	C
157	Kampala	Uganda	0.1	42.1	0.0	31.3	23.6	22.5	C
158	Alexandria	Egypt	0.0	49.5	0.0	88.7	10.8	30.6	C